

Reinforcement Learning



RUB

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EXPLORATION VS EXPLOITATION

Yingte and Youssef, 17th of June

Environments and Methods

ENVIRONMENTS

MiniGrid-Empty-8x8-v0
Open navigation task

MiniGrid-DoorKey-5x5-v0
Pick key → open door → goal

MiniGrid-DoorKey-8x8-v0
Same structure, larger map

MiniGrid-FourRooms-v0

MiniGrid-KeyCorridorS3R3-v0

VARIANTS

Baseline extrinsic reward only

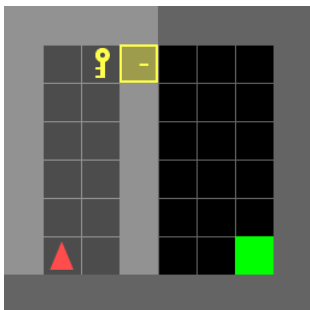
Baseline + noise extrinsic reward + observation noise

Intrinsic extrinsic + RND intrinsic reward

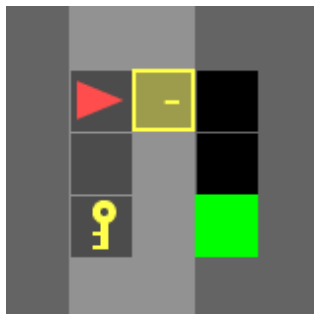
Intrinsic + noise RND intrinsic reward under noisy observations

5 envs * 4 variants * 3 seeds

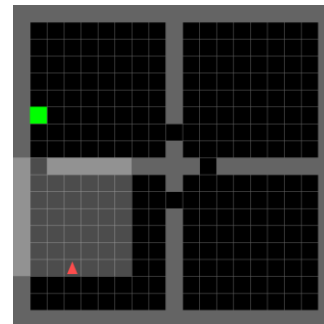
Environments



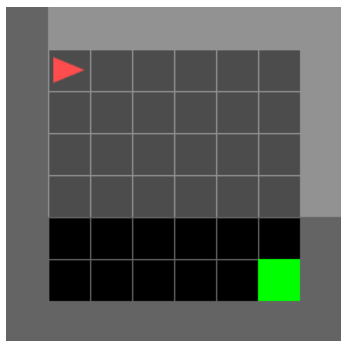
MiniGrid-DoorKey-8x8-v0



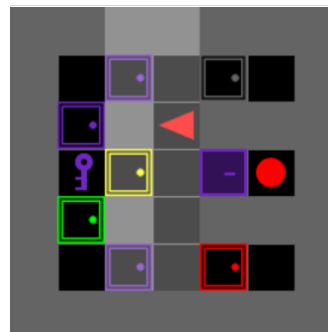
MiniGrid-DoorKey-5x5-v0



MiniGrid-FourRooms-v0

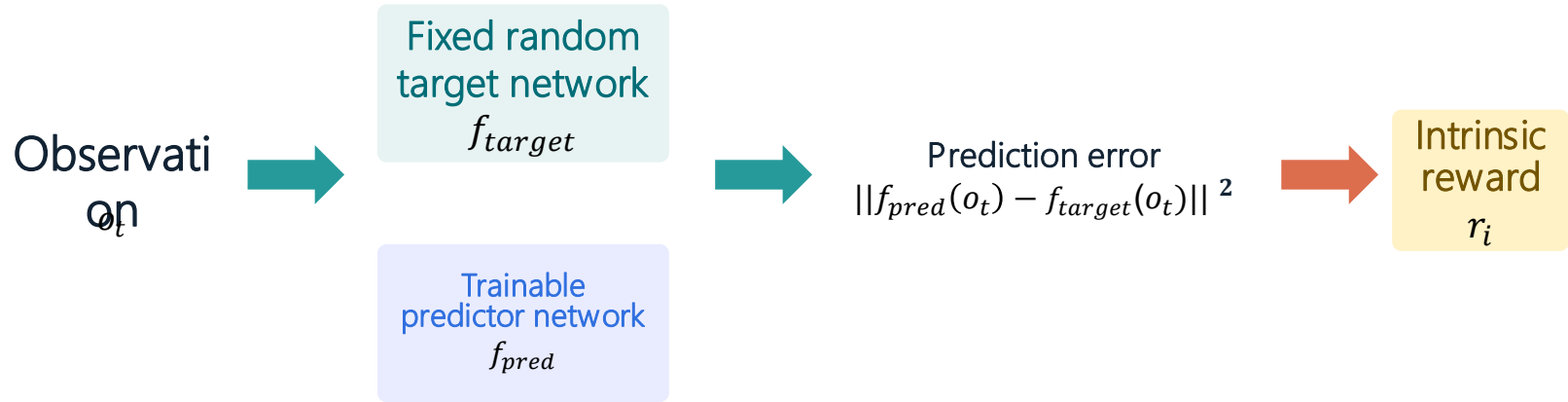


MiniGrid-Empty-8x8-v0



MiniGrid-KeyCorridorS3R3-v0

Intrinsic reward approach: Random Network Distillation



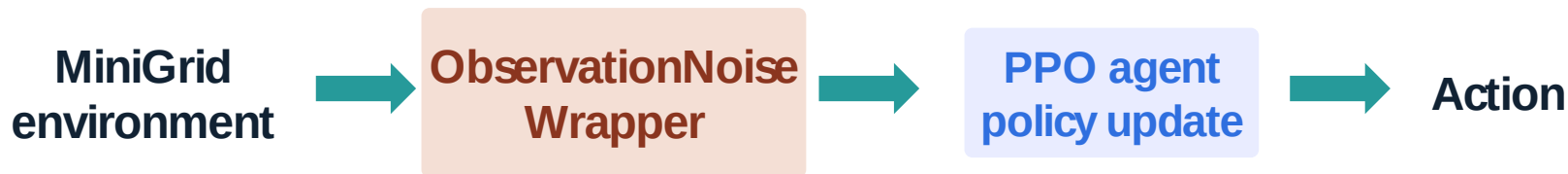
Total reward used for learning

$$r_{total} = r_{ext} + \beta \cdot r_{int}$$

- Unfamiliar observations have high prediction error.
- As the predictor learns them, the bonus decreases.
- The method encourages visiting novel states, not necessarily task-relevant states.

Noise implementation

Noise is implemented as an observation wrapper around the environment.



What changes?

The agent receives a corrupted/noisy observation. The environment transition dynamics and the external reward remain unchanged.

Why test it?

Noise tests robustness. It can also make RND overestimate novelty because the same state may look different to the predictor.

Evaluation metrics and presentation flow

1

Success rate

Did the agent reach the goal?

2

Mean reward

How much task reward was collected?

3

Episode length

Was the solution efficient or slow?

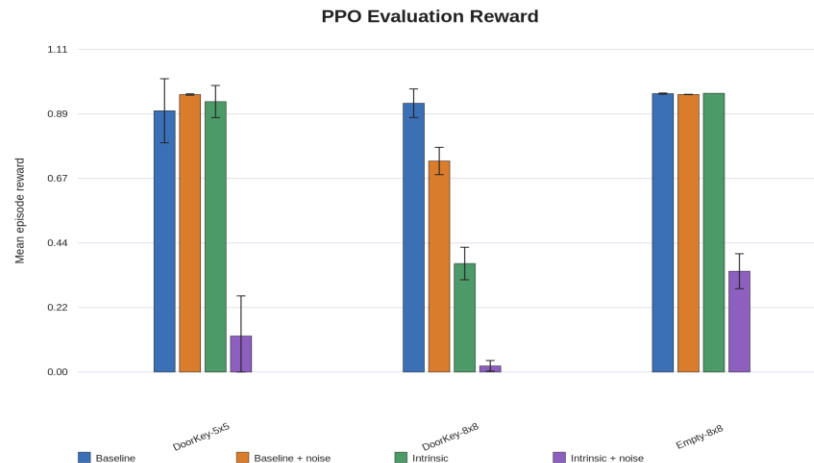
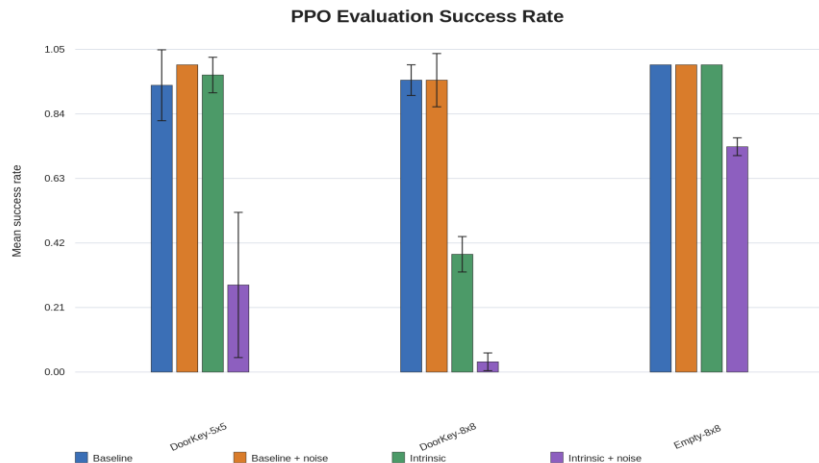
4

Success by seed

Was the behavior stable across runs?

Results: success rate and reward

The baseline is already very strong on the shown tasks; RND does not consistently improve it.

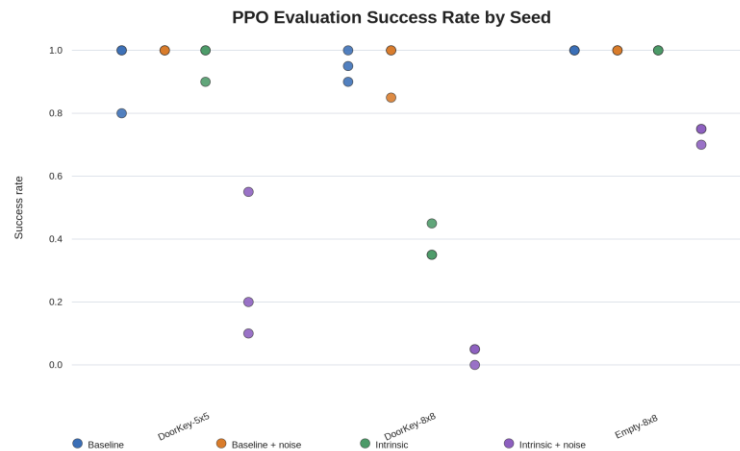
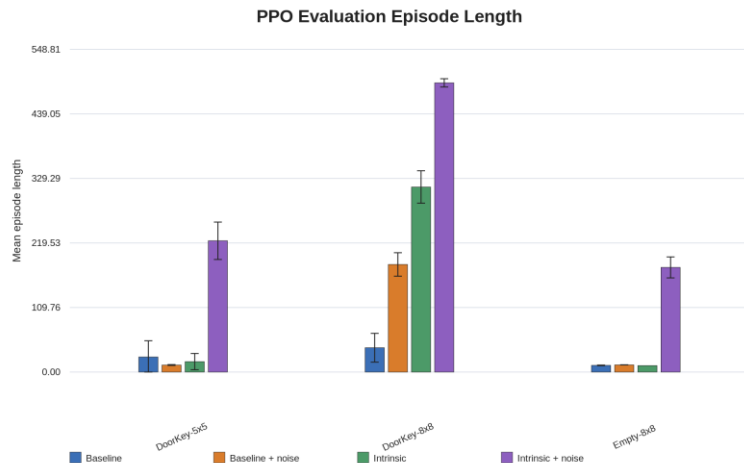


Key observations

- Empty-8x8 is nearly solved by all non-noisy variants.
- DoorKey-8x8: baseline success ≈ 0.95 , intrinsic success ≈ 0.38 .
- Intrinsic + noise performs worst, especially on DoorKey-8x8.

Results: efficiency and seed stability

Low success combined with long episodes suggests inefficient exploration.



Interpretation

- Intrinsic + noise leads to much longer episodes, especially in DoorKey-8x8.
- The seed plot shows that the failure is not only one unlucky run.
- RND can push exploration, but not always toward the actual sparse-reward objective.

Conclusion and next steps

The current evidence suggests that intrinsic reward must be tuned and evaluated carefully.

Current conclusion

RND did not outperform the baseline in the current PPO evaluation. The combination of RND and observation noise was especially problematic.

What will be explained in the discussion

- RND encourages exploration, but the bonus needs correct scaling.
- Noise can make the prediction error less meaningful.
- Harder environments may be better test cases for intrinsic rewards.
- Future ablations: β tuning, reward normalization, annealing, noise intensity, more seeds.

Next experiments: FourRooms / KeyCorridor S3R3

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